Critical Inquiry

Skeleton-Based Action Recognition in Classical Ballet

Using ST-GCN

Temporal Action Localization through Structured Skeleton Sequences

Supervisor:
Khoo Soo Guan

Team Members:
Liu Minghao, Wei Bo, Meng Xiangran

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Technical Roadmap: Skeleton-Based Action Recognition in Classical Ballet

PHASE 1: High-Fidelity Data & Feature Extraction



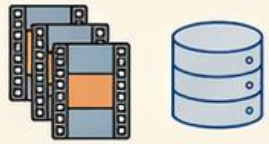
Primary Dancer Detection & Sampling

Extracting frames at 8 fps and using YOLOv8s to track the main performer.



65-Point Custom Skeleton

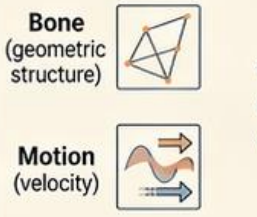
Reducing 133 MMPose keypoints to 65 (body, feet, hands) to eliminate facial noise



UJAnnChor Dataset Refinement

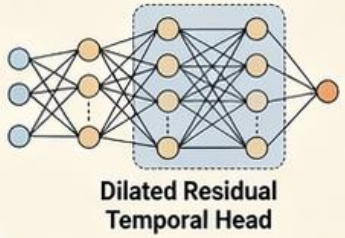
Processing 1,020 videos with 24,600 temporal annotations for 11 ballet action classes.

PHASE 2: Temporal Modeling & Evaluation



Dual-Stream ST-GCN Backbone

Fusing “Bone” and “Motion” streams for robust feature extraction.



Dilated Residual Temporal Head

Using a stronger temporal head to capture long-range transitions and improve foreground accuracy.



0.844 Segment F1 Score

Achieving stable action identification across diverse difficulty profiles and dense transitions.

Technology Stack

Component	Technology / Method	Primary Function
Detection Frontend	YOLOv8s	Real-time object detection and tracking
Pose Estimation	MMPose (COCO-WholeBody)	High-precision 133-keypoint extraction
Core AI Model	ST-GCN (with Dilated Head)	Temporal modeling and action segmentation

01 Part 1: Dataset and Task

Dataset: UJAnnChor

Public Ballet Dataset

Specifically designed for **classical ballet temporal action localization**, making it uniquely suited for this research domain.

1,020
Long Videos

Complete ballet performances with continuous action sequences

~24,600
Temporal Annotations

Precise start and end timestamps for each action segment

Label Space Configuration

11	+	1	=	12
Original Classes		Background		Total Classes

Data Split

Training Set 726

samples (90%)

Validation Set 81

samples (10%)

Key Advantages

- **Ballet-specific focus** — not generic actions
- **Temporal annotations** — provides action start/end times for precise localization

01 Part 1: Dataset and Task

Task: Frame Classification & Segmentation

1 Frame Classification

Assign a label to every frame in the video sequence, creating a frame-wise prediction map.

Example Output

Frame 1-45: Pirouette Frame 46-78: Grand Jete Frame 79-120: Background

2 Segmentation

Convert frame predictions into complete action segments with clear start and end boundaries.

Segment Output

- **Pirouette**
  Start: 0.0s | End: 5.6s | Duration: 5.6s
- **Grand Jete**
  Start: 5.7s | End: 9.8s | Duration: 4.1s

Pirouette



Grand Jete



Core Challenge

Accurate boundary localization — not just recognizing the action, but determining exactly when it starts and ends.

Ballet movements are **smooth and continuous**. Many categories differ only in **subtle changes** of leg angle, foot direction, or arm position.

The model may understand the action but still struggle to decide exactly when one action ends and another begins.

02 Part 2: The 12 Categories

The Label Space

11 Ballet Action Classes + 1 Background

1 **Pirouette** Spinning turn on one leg	2 **Grand Jete** Large horizontal jump	3 **Tour En Lair** Turn in the air
4 **Sissonne F** Jump from two feet	5 **Cabriole Derriere** Caper-like jump	6 **Courus** Running steps
7 **Waltz Step** Three-step movement	8 **Extension Second** Leg extension to side	9 **Ext Derriere** Leg extension to back

Label Distribution

Action Classes 91.7%

11 classes

Background 8.3%

1 class

Background Class

Critical addition — not every frame in a long video belongs to a target action.

Background captures:

- Pauses between movements
- Transition intervals
- Non-action periods

Prevents forcing non-action frames into action labels, improving model accuracy.

Note: Classes 10-11 are additional ballet movements completing the 11-action set.

Why Some Categories Are Harder

Easier Categories

Jump-based or large-extension actions have more distinctive skeleton patterns.

Grand Jete
Large horizontal leap — clear limb extension

Tour En Lair
Aerial rotation — distinct vertical movement

Courus
Running steps — repetitive motion pattern

Why easier: Large body displacement and dramatic pose changes create clear skeleton signatures.

Harder Categories

Similar skeleton patterns and tendency to appear in rapid succession.

Pirouette
Spinning turn — subtle body position changes

Extension Second
Leg extension to side — minimal skeleton variation

Ext Derriere
Leg extension to back — similar to Extension Second

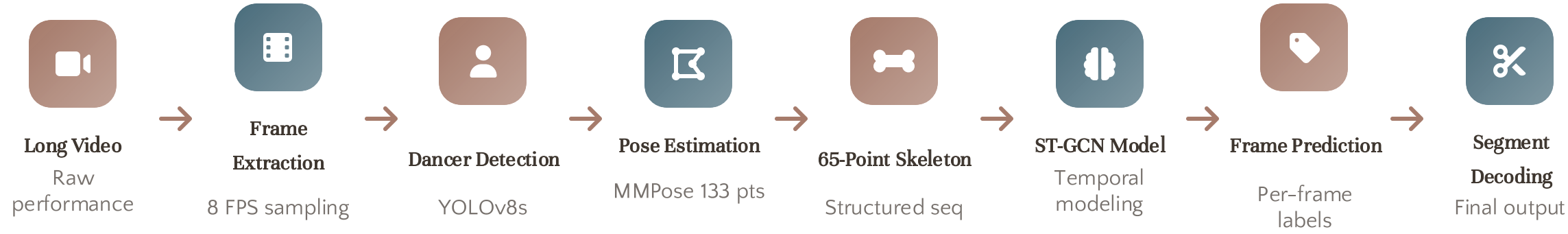
Why harder: Model must solve **fine-grained classification** + **precise temporal separation** simultaneously.

Dual Challenge: This simultaneous requirement for fine-grained discrimination and precise boundary localization is one of the **main reasons why ballet action recognition is challenging**.

03 Part 3: Overall Pipeline

Pipeline Overview

Complete Processing Flow



Visual Input

Raw performance videos with complete ballet sequences

Structured Data

Skeleton sequences encoding body structure and movement

Temporal Output

Frame labels and action segments with precise boundaries

Why Skeleton-Driven Pipeline

RGB Approach Limitations

Pure RGB end-to-end approaches capture all visual information — including irrelevant noise.

Costume Color
Tutu color, lighting variations — not essential for action recognition

Stage Background
Scenery, props, curtains — distract from dancer movement

Lighting Conditions
Spotlight changes, shadows — introduce visual inconsistency

Skeleton Approach Advantages

Preserves essential cues while reducing visual noise from irrelevant content.

Body Structure
Joint positions, limb relationships — core information

Limb Extension
Leg angles, arm positions — critical for ballet

Foot Placement
Pointe work, foot direction — distinguishes actions

Movement Transition
How poses evolve over time — temporal dynamics

Key Insight

Although the representation is more abstract, it is actually more aligned with the nature of ballet analysis

Early Pipeline: Frame Extraction & Detection

1 Frame Extraction

Uniform sampling at 8 frames per second — preserves motion detail while maintaining computational efficiency.

8 FPS
Sampling Rate

Balance
Detail vs Efficiency

Why 8 FPS? Captures enough temporal information for ballet movements while keeping later stages computationally manageable.

2 Primary Dancer Detection

Selection Criteria

- Bounding box area
- Image centrality
- IoU continuity
- Bottom-region filter

Middle Pipeline: Pose Estimation & Skeleton

3 Whole-Body Pose Estimation

MMPose with COCO-WholeBody setting extracts 133 keypoints — far more than simple 17-joint skeletons.

Why not 17 joints?

Ballet depends on **detailed cues** that 17-joint skeletons miss:

Hand Shape	Foot Position	Leg Extension
Finger positions	Pointe, flexion	Height, angle

4 65-Point Skeleton Construction

Task-specific design — selective keypoint retention from 133 to 65 points.

Keypoint Distribution 133 → 65

Body points	17
Foot points	6
Left hand points	21
Right hand points	21
Face landmarks	Removed

Design rationale: Face landmarks add noise; hands and feet carry critical information for ballet action recongition.

Late Pipeline: Input Representation & Training

Input Representation

Three representations compared: bone, motion, and dual-stream (bone + motion).

Bone Stream

Main Signal

Joint-to-joint vectors. Captures **spatial body structure** — strongest standalone signal.

Motion Stream

Auxiliary

Temporal displacement vectors. Captures **movement dynamics** — works better as auxiliary.

Dual-Stream

Combined

Bone + Motion fusion. Leverages **complementary information** for robust feature extraction.

Finding: Bone is the **stronger main signal**; motion works better as auxiliary stream.

Training Strategies

Addressing class imbalance and difficult segments *through targeted sampling and loss reweighting.*

Foreground Boosting

Addresses **action vs. background frame imbalance** by up-weighting foreground samples during training — prevents the dominant background class from drowning out action signals.

Short-Segment Boosting

Increases attention to **brief / difficult action segments** that are easily missed — critical for precise boundary localization in dense transitions.

Combined: ***shortseg_ft*** — fine-tunes with both strategies active to balance overall recall and rare-class accuracy.

Temporal Modeling: Inside ST-GCN

ST-GCN = **Spatial** + **Temporal** + **Graph Convolutional Network** — *a backbone that reasons over the skeleton as a graph, evolving through time.*

S

Spatial

Graph structure over joints

Builds a **skeletal graph** where 65 keypoints are nodes and anatomical bones are edges — the model reasons about **body-part relationships** (hip ↔ knee ↔ ankle).

T

Temporal

Time-axis convolutions

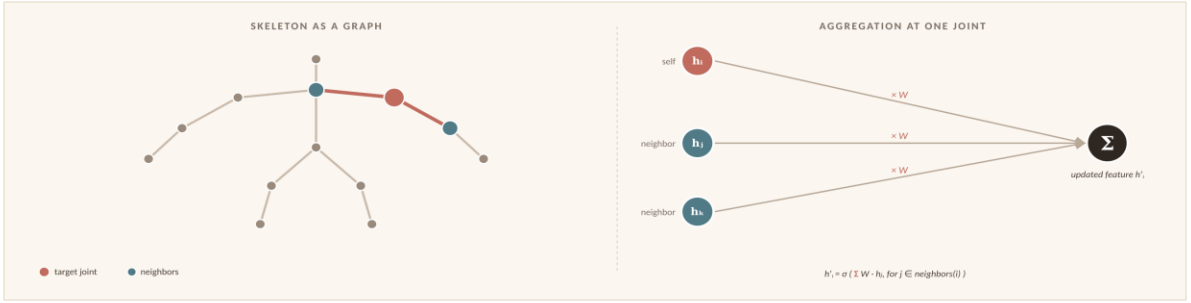
Applies **1-D convolutions along time** at each joint — tracks a **pirouette spin** or **grand-jeté arc** as a trajectory, not isolated poses.

GCN

Graph Convolutional Network

Aggregate neighbor features over the skeleton graph

For each joint, GCN **pulls features from its connected neighbors** and combines them with learned weights — like CNN, on a graph.



Why ST-GCN for Ballet

Ballet is inherently **spatial and temporal**, we should not only pay attention to the place of the joints but also the movement of them, And since skeleton points are connected to neighbors and the relations should be focused on.

Evaluation: Frame-Level Metrics

Two-level evaluation system. Frame-level metrics assess per-frame classification accuracy before segment decoding.

Overall Accuracy

Standard metric: proportion of correctly classified frames across all classes.

Accuracy = Correct / Total Frames

Limitation: Can appear overly optimistic due to common background frames.

Foreground Accuracy

More important metric: evaluates model only on action frames (excluding background).

FG Acc = Correct Actions / Total Action Frames

Why it matters: Reveals true action recognition performance without background bias.

Foreground Macro-F1

Class-balanced metric: reveals whether model performs well across all categories or just easier ones.

Macro-F1 = Mean(F1 per class)

Reveals: Performance balance — does model favor easy categories?

Metric Selection Rationale

Frame-level metrics provide **granular understanding** of classification performance before segment-level evaluation. Foreground-focused metrics are essential due to background class dominance.

Evaluation: Segment-Level Metrics

More important for temporal localization: convert frame predictions into segments and evaluate boundary precision.

Segment F1 at 3 IoU Thresholds

Intersection over Union (IoU) measures overlap between predicted and ground-truth segments.

F1 @ IoU 0.3
Lenient

Coarse localization acceptable

F1 @ IoU 0.5
Main Metric

Balance between coarse and strict

F1 @ IoU 0.7
Strict

Precise boundary matching required

Boundary Error Metrics

Start Error
|Predicted Start - GT Start|

End Error
|Predicted End - GT End|

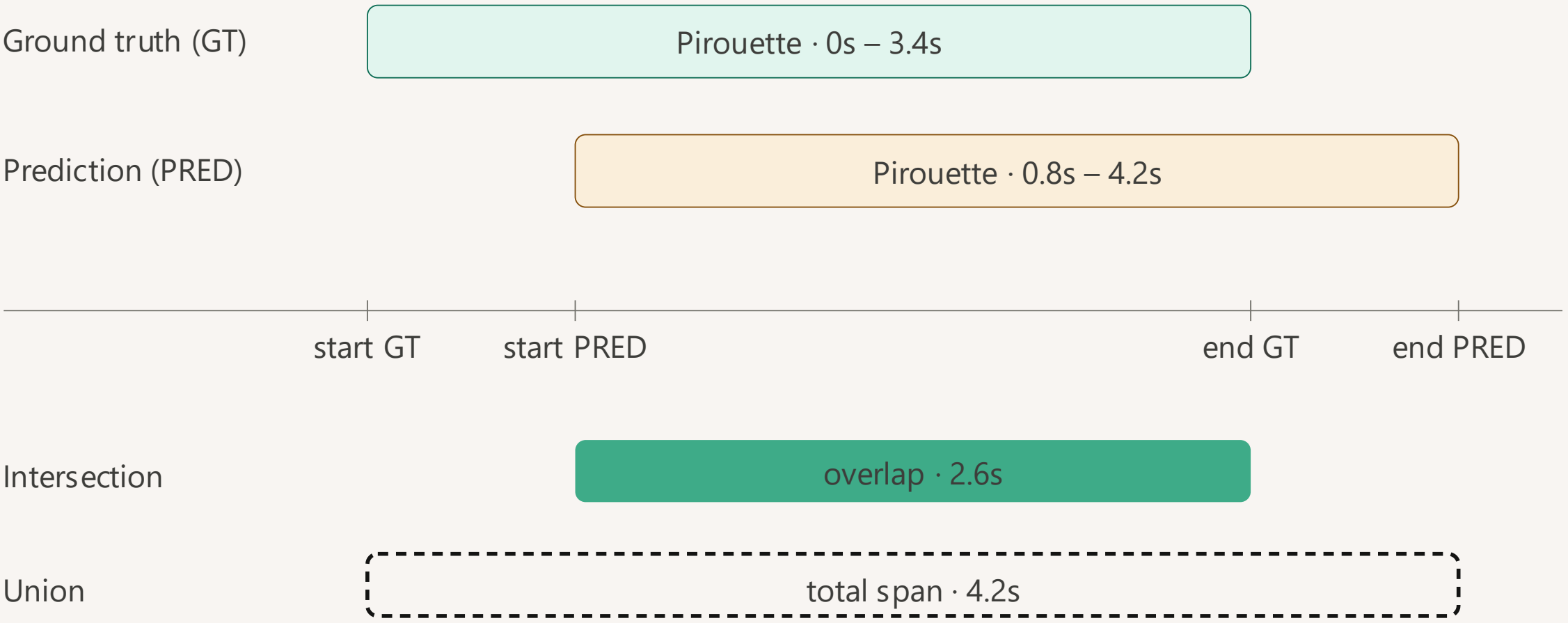
Duration Error
|Predicted Duration - GT Duration|

Predicted Segment Count Analysis

Difference between predicted and ground-truth segment counts reveals:

- Missing actions (under-detection)
- Over-merging neighboring actions
- Splitting segments too aggressively

IoU = how much the predicted time interval overlaps the real one



IoU = 2.6 / 4.2 ≈ 0.62

→ the two intervals mostly agree, but not perfectly

How strict is each IoU threshold?

Teal = ground truth · Amber = prediction · Green = overlap · Dashed = union

IoU ≥ 0.3 — loose, "roughly in the right place"

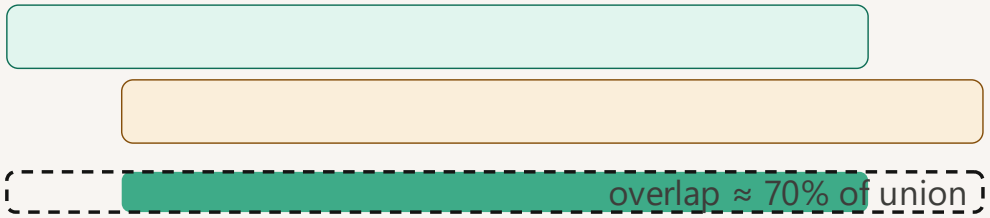


IoU ≥ 0.5 — standard, "overlap more than disagree" ⭐ main metric



It's a reasonable balance between rough localization and boundary precision.

IoU ≥ 0.7 — strict, "near-perfect boundary alignment"



Higher IoU = greater overlap required = tighter boundary precision

Overall Results

Three model lines show different strengths — complexity does not automatically mean better segmentation.

Single Bone

Strength

Most stable segment-level performance

Average F1@0.5 **0.844**

Fixed 5-video set

- Clean segmentation with minimal over-merging

Dual + Strong Head

Strength

Best foreground accuracy

Foreground Acc **0.889**

Highest sensitivity

- Most sensitive to foreground actions

Dual Bone + Motion

Strength

Best foreground macro-F1

Macro-F1 **0.787**

Most balanced

- Most balanced across categories

Counterintuitive Insight: : More complex ≠ better

A more complex model does not automatically lead to better segmentation. Stronger temporal head may detect action regions more actively, but can also over-merge neighboring short segments.

High-Accuracy Case Analysis

Example Videos

FPM0014 — Excellent

Clear boundaries and distinctive skeleton patterns

ACM0014 — Excellent

Well-separated action segments

Performance Result

TCM0014 - Single Bone Model

F1@0.5 = 1.0

Perfect segmentation

Model can **both classify actions correctly and segment cleanly** when action structure is sufficiently clear.

Success Factors

- ✓ **Clearer boundaries** between actions
- ✓ **Distinctive skeleton patterns** for each class
- ✓ **Well-separated** action segments

Conclusion

This demonstrates that the **skeleton-driven pipeline is fundamentally effective** when:

- Action structure is **sufficiently clear**
- Actions are **well separated** in time
- Skeleton patterns are **distinctive**

Low-Accuracy Case & Conclusions

Challenging Example

NUP0006

Busy stage, crowded scene, detection box and skeleton joints failing

Challenge Characteristics

- Categories are **very similar in skeleton space**
- **Rapid succession** of different actions
- Unclear **boundary transitions**

Result: All three of our models struggle on this video

Final Conclusion

Our work builds a **complete and interpretable ballet action recognition pipeline**, demonstrating that **skeleton-based modeling is effective** for this task. The most important next step is better **boundary-aware learning**.

Main Bottleneck

The harder problem is **no longer action detection**:

Fine-Grained Discrimination

Distinguishing between **similar skeleton patterns** in different action classes

Precise Boundary Localization

Determining exactly when **similar actions switch quickly**

Next Steps

Not simply a **stronger classifier**, but:

Better Boundary-Aware Learning

Ballet is one of the most refined forms of human expression.
Getting AI ***one step closer to actually seeing it*** *—*
We think that's a pretty cool thing to work on.

Thank You!